

RAPM: Retrieval-Augmented Pragmatic Mapper for Speech-to-Speech Translation

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Abstract

This paper presents RAPM (Retrieval-Augmented Pragmatic Mapper), a non-parametric system for the Interspeech 2026 TOPI Challenge. We investigate whether retrieving cross-lingual exemplars from a datastore guides Speech-to-Speech Translation (S2ST) prosodic mapping more effectively than parametric models. Our best configuration retrieves neighbors in a 103-dim pragmatically-salient subspace, achieving 0.8331 cosine similarity on the blind test set (± 0.002 over 10 runs), below the MLP baseline (0.8574). We report two statistically validated findings. First, retrieval in the full 1024-dim HuBERT space is significantly degraded by speaker identity interference on unseen speakers ($p < 0.05$); projecting queries into a curated subspace mitigates this effect. Second, the parametric fusion component yields negligible gains ($+0.0004 \pm 0.002$) with only $\sim 3k$ training pairs, revealing a data-efficiency bottleneck in hybrid architectures at this scale.

Index Terms: cross-lingual prosody transfer, retrieval-augmented generation, speech-to-speech translation, TOPI challenge

1. Introduction

Speech-to-speech translation (S2ST) systems have made substantial progress in preserving semantic content and speaker timbre, driven by advances in both cascaded and end-to-end architectures [1, 2, ?]. Yet current systems primarily optimize for linguistic accuracy and acoustic naturalness, leaving the *paralinguistic* dimension—how something is said, as opposed to what is said—largely unaddressed. In conversational speech, a speaker’s stance, attitude, and pragmatic goals—collectively termed *illocutionary force* [3]—are conveyed through prosodic cues such as pitch contours, rhythm, and pausing patterns. These cues are known to carry communicative weight comparable to lexical content in many interactive settings [4]. Self-supervised speech representations like HuBERT [5] capture rich acoustic information, but they also heavily encode speaker identity and channel characteristics [6], complicating the extraction of speaker-independent pragmatic structure.

The Interspeech 2026 Transfer of Pragmatic Intent (TOPI) Challenge targets this gap directly. It uses the Dialogs Reenacted Across Languages (DRAL) corpus [7, 8], in which bilingual speakers reenact matched utterance pairs aligned by communicative intent rather than literal translation. The task is to map a 1024-dimensional English HuBERT feature vector to a 101-dimensional Spanish prosodic representation—a many-to-many problem, since identical text can carry different pragmatic meanings depending on prosody, and cross-lingual prosodic

conventions are rarely isomorphic [9]. The official baseline learns this mapping with a Multi-Layer Perceptron (MLP), which optimizes a global regression objective over the training distribution.

A known limitation of parametric regression in such settings is its tendency toward the conditional mean, which can underrepresent the diverse prosodic modes present in the tail of the distribution. Retrieval-based methods offer a complementary inductive bias: rather than compressing the training distribution into fixed network weights, they preserve individual exemplars and construct predictions by locally interpolating among the most relevant ones. This paradigm has shown promise in language modeling [10], open-domain question answering [11], and voice conversion [12], where it improves generalization without additional parametric training. Motivated by these results, we propose **RAPM** (Retrieval-Augmented Pragmatic Mapper), which retrieves cross-lingual exemplars from a datastore to construct a prosodic prior for each input. A practical advantage of this design is transparency: unlike parametric baselines, every prediction can be traced to specific training exemplars, facilitating interpretable error diagnosis.

Our working assumption is that in a tightly matched corpus like DRAL, an English query can index a relevant Spanish prosodic prototype because the underlying pragmatic structure is shared across languages. This assumption holds most strongly when the query speaker is represented in the datastore; as we show below, it weakens for unseen speakers. Our contribution is primarily diagnostic. Through controlled experiments with statistical validation (std. ± 0.002 over 10 runs), we identify two findings: (1) retrieval in the full 1024-dim HuBERT space is significantly degraded by speaker-identity interference on unseen speakers ($p < 0.05$, paired t -test), and a curated 103-dim pragmatic feature subset mitigates this effect; (2) the parametric fusion component provides negligible correction given only $\sim 3k$ training pairs, revealing a data-efficiency bottleneck in hybrid retrieval-parametric architectures at this scale.

2. System Description

2.1. Task and Feature Dimensions

The task involves mapping an English HuBERT [5] feature vector to a Spanish prosodic representation. Three distinct dimensionalities appear throughout this work, summarized in Table 1.

The 103 and 101 dimensions are not learned projections but *hard index selections*: subsets of HuBERT dimensions identified by Ward et al. [14] through correlation with human pragmatic similarity judgments. We utilize the official dataset [7, 8] consisting of 2,893 aligned utterance pairs.

** indicates the corresponding author.

Table 1: *Feature dimensions used in RAPM.*

Dim	Source	Role in RAPM
1024	HuBERT output	Full English input $\mathbf{x}_{en}$
103	english_winners [13]	Config B retrieval keys
101	spanish_winners [13]	Target output $\mathbf{y}$

2.2. The Retrieval Module

Our approach is inspired by nearest-neighbor language models [10] (see Figure 1). We construct a key-value datastore $\mathcal{D} = \{(\mathbf{k}_i, \mathbf{v}_i)\}_{i=1}^N$ from the training set. Given a query $\mathbf{q}$, we compute cosine similarity $s_j = \cos(\mathbf{q}, \mathbf{k}_j)$ against all keys, select the top- K neighbors, and aggregate their values using a temperature-scaled softmax:

$$\hat{\mathbf{y}}_{ret} = \sum_{j \in \text{Top-}K} \frac{\exp(s_j/\tau)}{\sum_{m \in \text{Top-}K} \exp(s_m/\tau)} \mathbf{v}_j \quad (1)$$

2.3. Retrieval Configurations

To isolate the effect of feature dimensionality on generalization, we investigate two retrieval workflows:

Config A: High-Res (1024-dim). This baseline configuration utilizes the full representational capacity of HuBERT. Both datastore keys $\mathbf{k}$ and values $\mathbf{v}$ remain 1024-dimensional vectors. The retrieved aggregate is projected to the 101-dim target space (via `spanish_winners` indices) only *after* aggregation, immediately prior to the Fusion Network.

Config B: Pragmatic Subspace (103-dim). High-dimensional speech embeddings often entangle linguistic content with paralinguistic *nuisance variables* like speaker timbre. To mitigate this, we project both input queries $\mathbf{x}_{en}$ and datastore keys $\mathbf{k}$ onto the 103 `english_winners` indices [13]. This hard selection acts as an *information bottleneck* [15], filtering out identity-dominated dimensions while retaining those correlated with pragmatic intent. Crucially, while the *search* occurs in this noise-robust subspace, we retrieve the associated full 1024-dim values $\mathbf{v}$ (subsequently projected to 101-dim), allowing the system to construct a rich prosodic prior from pragmatically matched exemplars.

2.4. The Fusion Network

The fusion network learns a residual correction on top of the retrieved prior, adopting principles from residual learning [16] (Figure 2). The residual design reflects the assumption that the retrieval module already provides a reasonable initial estimate, so the network need only learn a small correction $\Delta\mathbf{y}$ rather than the full mapping. This inductive bias is beneficial when retrieval quality is high, but may limit the network’s corrective capacity when retrievals are poor (e.g., for unseen speakers).

For both Config A and Config B, the retrieved values are projected to 101 dimensions (via `spanish_winners`) before entering the Fusion Network, so the input dimensionality is consistently $1024 + 101 = 1125$:

$$\mathbf{y}_{pred} = \hat{\mathbf{y}}_{ret}^{(101)} + \mathcal{F}_\theta(\text{concat}[\mathbf{x}_{en}, \hat{\mathbf{y}}_{ret}^{(101)}]) \quad (2)$$

Note that while Config B uses 103-dim queries for the retrieval phase, the Fusion Network consistently receives the original 1024-dim $\mathbf{x}_{en}$ to preserve the full source context (as detailed in Figure 2). The MLP employs a standard feed-forward architecture: $[1125 \rightarrow 256 \rightarrow 128 \rightarrow 101]$, interspersed with Layer-

Norm and GELU activations. This funnel-like design acts as a structural regularizer; by progressively reducing the dimensionality of the concatenated representations, the network is encouraged to discard redundant acoustic noise. It focuses its capacity solely on extracting the salient pragmatic cues necessary to output the precise residual correction $\Delta\mathbf{y}$.

3. Experimental Setup

3.1. Data Configuration

We used the standard train/test split from the official baseline repository. The training set contains 2,893 aligned English–Spanish utterance pairs from the DRAL corpus. To enable early stopping without contaminating held-out evaluation, we further split the training set into a 90% training subset (2,604 pairs) and a 10% validation subset (289 pairs), stratified by speaker to preserve the per-speaker distribution. The full training set (2,893 pairs) serves as the retrieval datastore at inference time.

- **Internal Split (Seen):** The standard baseline test split. This split shares speakers with the training set, so it primarily measures in-domain fitting quality. Hyperparameters K and τ were tuned on this split.
- **Official Challenge Set (Unseen):** The blind test set released by the organizers. Four of five speakers in this set are absent from the training corpus, making it a stringent test of speaker-independent generalization.

3.2. Implementation Details

All experiments were conducted on a single NVIDIA RTX 3090 GPU. Each configuration was run 10 times with different random seeds to obtain mean and standard deviation; we report seed 42 as the representative submission. The Fusion Network was trained for up to 100 epochs with AdamW (learning rate 1×10^{-3} , weight decay 1×10^{-4} , batch size 32) and Cosine Embedding Loss, with early stopping (patience = 10) monitored on the held-out validation subset. Retrieval was performed via ChromaDB with cosine distance on GPU. The search grid for K and τ is detailed in Table 5.

4. Results

4.1. Main Results

Table 2 summarizes performance across configurations. All values are means over 10 runs; standard deviations are ± 0.002 throughout. Statistical significance is assessed with paired t -tests across runs.

Table 2: *Main results (mean over 10 runs, ± 0.002). †: significantly different from the corresponding Config A entry ($p < 0.05$, paired t -test). Gain = absolute improvement from the Fusion Network.*

System	Ret. Dim	Internal (Seen)		Official (Unseen)	
		Cosine	Gain	Cosine	Gain
Baseline MLP	–	0.8732	–	0.8574	–
<i>Config A: Full-Space</i>					
Pure Retrieval	1024	0.8722	–	0.8286	–
+ Fusion	1024	0.8742	+0.0020	0.8290	+0.0004
<i>Config B: Subspace</i>					
Pure Retrieval	103	0.8730	–	0.8318†	–
+ Fusion	103	0.8741	+0.0011	0.8331†	+0.0013

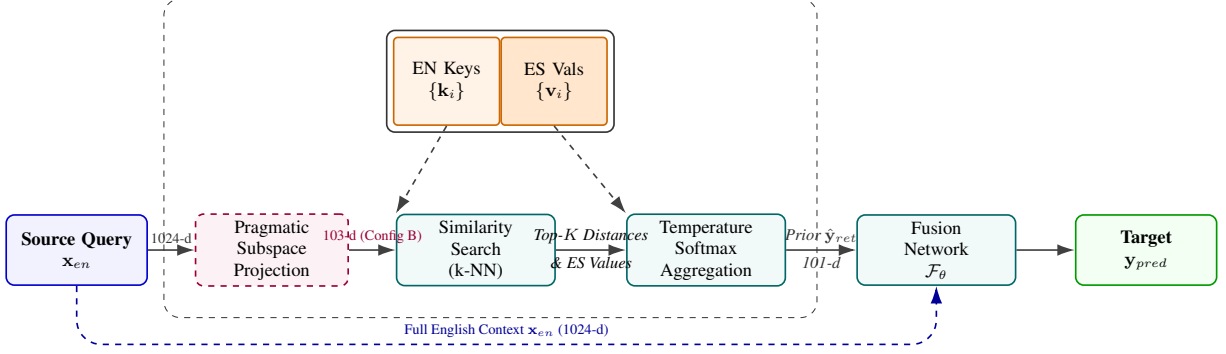


Figure 1: *RAPM System Architecture: The Pragmatic Subspace Projection filters out identity noise (Config B). The system matches English keys but retrieves the associated Spanish prosodic values, generating a cross-lingual prior $\hat{y}_{ret}$ which is subsequently refined by the Fusion Network.*

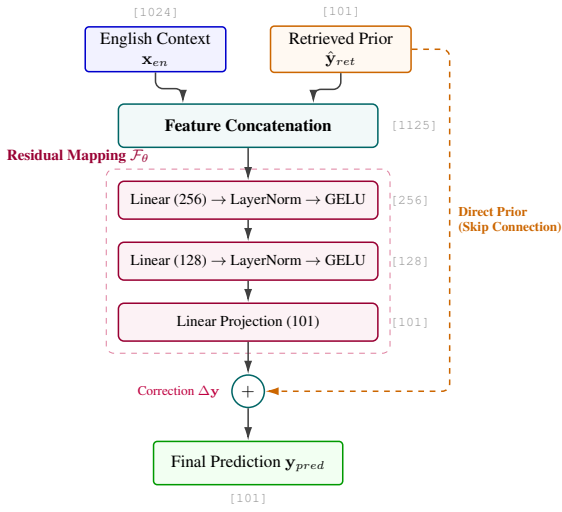


Figure 2: *Fusion Network details: The network explicitly models the non-linear residual correction F_θ . It maps the concatenated 1125-dim feature through a progressive dimensionality reduction to learn localized prosodic refinements.*

On seen speakers, both configurations match the baseline (0.874 vs. 0.873). The gap emerges on unseen speakers: our best result (0.8331) falls short of the baseline (0.8574) by 0.024. We analyze the sources of this gap below.

Speaker identity interference. Config B (103-dim) significantly outperforms Config A (1024-dim) on unseen speakers: 0.8318 vs. 0.8286 for pure retrieval ($p < 0.05$). Self-supervised representations like HuBERT encode speaker identity heavily [6], so nearest-neighbor search in the full space tends to match acoustic profile rather than pragmatic intent—an effect also observed in kNN-based voice conversion [12]. The curated 103-dim subspace filters out identity-correlated dimensions, yielding retrievals that better align with the underlying communicative intent.

Limited fusion gains. The Fusion Network adds +0.0013 for Config B on unseen speakers ($p < 0.05$), but only +0.0004 for Config A ($p > 0.1$, not significant). With 2,893 training pairs spanning $\mathbb{R}^{1024}$, the residual correction network lacks sufficient manifold coverage to generalize corrections to unseen speaker regions. This suggests that hybrid retrieval-parametric archi-

tectures require substantially larger datastores to surpass purely parametric baselines in this task.

4.2. Ablation: Feature Space Selection

Table 3 compares retrieval design choices on the internal split.

Table 3: *Ablation on internal split (seen speakers). Δ : difference from the reference configuration (softmax, $K=70$, $\tau=0.04$).*

Setting	Cosine	Δ
<i>Aggregation</i>		
Hard Top-1	0.7859	−0.0883
Top-Truncated ($K=50$)	0.8734	−0.0008
Softmax ($K=70$, $\tau=0.04$)	0.8742	Ref.
<i>Feature Space</i>		
PCA (101-dim)	0.4456	−0.4286
Winners (103-dim)	0.8741	−0.0001

Two observations stand out. First, soft aggregation over $K=70$ neighbors is essential; hard top-1 retrieval drops cosine by 0.088, confirming that no single exemplar adequately represents the target prosody.

Second, PCA reduction to 101 dimensions collapses performance to 0.4456. PCA retains directions of maximum variance, which in HuBERT space correspond primarily to speaker identity and channel characteristics rather than pragmatic content. The curated winners indices, selected by correlation with human pragmatic similarity judgments [14], preserve performance at 0.8741. This confirms that unsupervised dimensionality reduction is not a viable substitute for task-specific feature selection in this setting.

4.3. Ablation on Fusion Inputs: Representational Trade-offs

As observed earlier, the Fusion Network provides negligible gains over pure retrieval on unseen speakers. We hypothesized that feeding the full 1024-dim English HuBERT vector into the Fusion Network re-introduces source speaker acoustic traits, increasing the risk of overfitting to seen speakers.

To investigate this, we conducted an ablation study on the internal test split (seen speakers) to observe the network’s behavior when its input is strictly limited to speaker-independent

representations. We evaluated two variants based on Config B ($K=70$, $\tau=0.04$): (1) **Consistent Subspace Fusion (204-dim)**, which concatenates only the 103-dim pragmatic subset and the 101-dim retrieved prior; (2) **Prior-Only Refinement (101-dim)**, which provides only the retrieved prior without any English source context.

Table 4: *Fusion input ablation on the internal split (seen speakers). Differences smaller than 0.002 are not statistically significant.*

Fusion Network Input	Input Dim	Cosine (± 0.002)
Original Fusion (1024 + 101)	1125	0.8741
Pure Retrieval Baseline	–	0.8730
Prior-Only (101)	101	0.8683
Consistent Subspace (103 + 101)	204	0.8660

As shown in Table 4, removing the high-dimensional source context resulted in a performance decrease on seen speakers (from 0.8741 to 0.8660), yielding results slightly below the pure non-parametric retrieval baseline (0.8730).

This outcome highlights a fundamental representational trade-off for hybrid retrieval-parametric systems in low-resource S2ST:

- **Source Feature Over-reliance:** Providing the full 1024-dim context allows the network to leverage rich acoustic cues, which improves performance on seen speakers but makes the model susceptible to overfitting the source speaker’s timbre, thereby limiting generalization to unseen speakers.
- **Contextual Sparsity:** Restricting the input to a curated pragmatic subspace (e.g., 103-dim) effectively filters out identity-related variance. However, this compression removes fine-grained phonetic details that may be necessary for computing precise residual corrections ($\Delta\mathbf{y}$), causing the parametric module to perform marginally worse than pure retrieval.

Consequently, under the current data scale, a well-calibrated pure retrieval module operating in a pragmatic subspace remains a competitive and robust baseline compared to introducing a residual parametric layer.

4.4. Hyperparameter Sensitivity

Table 5 shows the grid search over K and τ (pure retrieval, internal split, Config A).

Table 5: *Hyperparameter grid search (internal split). Values within ± 0.002 are statistically indistinguishable.*

K	Cosine	τ	Cosine
1	0.786	0.01	0.844
10	0.864	0.03	0.871
30	0.871	0.04	0.872
70	0.872	0.07	0.872
90	0.874	0.10	0.871

Performance rises steeply from $K=1$ to $K=30$, then plateaus. $K=70$ and $K=90$ are within the ± 0.002 noise floor; we select $K=70$ as the smaller value reduces the risk of including irrelevant neighbors for unseen speakers. Temperature is

similarly stable between 0.04 and 0.07; we use $\tau=0.04$, which concentrates weight on the closest matches. We also explored Optimal Transport-based weighting but found softmax aggregation more stable with comparable accuracy.

4.5. Qualitative Observations

A practical advantage of retrieval-based prediction is traceability. For any output, the retrieved neighbors can be inspected to diagnose failure modes. During development, we observed that in the 1024-dim space, top neighbors for a given query were dominated by utterances from the same speaker (when present in the datastore), regardless of dialog act. In the 103-dim subspace, retrieved neighbors came from multiple speakers but shared the pragmatic function of the query (e.g., empathetic backchannels, hesitant pauses). This pattern is consistent with the quantitative finding that subspace retrieval generalizes better to unseen speakers.

5. Discussion & Conclusion

Although RAPM (0.8331) does not surpass the MLP baseline (0.8574) on unseen speakers, our ablation study reveals a fundamental *representational trade-off* in hybrid S2ST architectures (Section 4.3).

Richness vs. Robustness Trade-off. Parametric models benefit from global weight sharing, implicitly learning speaker-invariant mappings. In contrast, our hybrid system faces a dilemma: supplying the full 1024-dim source context provides necessary linguistic detail but induces overfitting to the source speaker’s acoustic profile. Conversely, restricting input to a curated pragmatic subspace (103-dim) mitigates identity interference but creates *contextual sparsity*, depriving the fusion network of the fine-grained cues needed for precise residual corrections.

Data Density and Retrieval Dynamics. The performance gap also highlights a scale bottleneck. With only 2,893 exemplars, setting $K=70$ aggregates $\sim 2.4\%$ of the entire datastore for each prediction. This behavior resembles global kernel smoothing rather than the sparse, selective retrieval observed in billion-scale text LMs [10]. We hypothesize that bilingual data augmentation is required to shift the system into a high-density regime where retrieval offers a genuine non-parametric advantage.

Conclusion. We conclude that in low-resource regimes, a pure non-parametric retrieval module operating in a pragmatic subspace is structurally more robust than hybrid parametric fusion. While the current retrieval approach offers superior interpretability—enabling distinct error diagnosis—future improvements must move beyond hard feature selection. The path forward lies in learning continuous, disentangled representations [17] that preserve linguistic context while explicitly suppressing speaker identity.

6. Reproducibility

The code and checkpoints will be made publicly available after the peer review process.

7. Generative AI Use Disclosure

The authors acknowledge the use of Google’s Gemini 3.0 Pro Preview and Anthropic’s Claude Opus 4.5 Thinking for language polishing and assistance in code development. All final content, including experimental design, results analysis, and conclusions, was verified and approved by the authors.

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